

Fiber-Optic Cables — Guided Notes

Unit 1.3.2 | CompTIA Network+ N10-009 Obj. 1.3 | N10-008 1.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Fiber-optic cable transmits data as pulses of _____ instead of electrical signals. This makes it immune to _____ and safe to run between buildings without risk of _____.

Question 2

Multimode fiber has a _____ (larger/smaller) core than single-mode. Its light source is a _____, which is cheaper than the _____ used in single-mode. The main distance limitation of multimode is caused by _____.

Question 3

Single-mode fiber has a core of approximately _____ microns and supports runs of up to _____ km without a repeater. Its jacket color is _____. Multimode jacket colors are _____ or _____.

Question 4

Match each fiber connector to its description:

SC → _____

LC → _____

ST → _____

Question 5

Fiber connections always come in pairs because one strand _____ and the other _____. Swapping these two strands is a common mistake that mixes up _____ and _____.

Question 6

UPC connectors have a _____ housing. APC connectors have a _____ housing and polish the endface at a _____ angle. These two types are physically _____ and must never be mated.

Question 7

Exceeding a fiber cable's minimum _____ causes macrobending loss. Contaminated fiber endfaces must be cleaned with _____ before every connection. Unused connectors and ports should always be _____.

Question 8 — Real World Application

REAL WORLD: A backbone fiber link between two wiring closets starts throwing CRC errors and negotiates at a lower speed. Both ends use LC connectors with blue boots. Walk through your troubleshooting steps in order.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-3-2-fiber-optic-cables/1-3-2-fiber-optic-cables-answer-key.pdf
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